

Scalable Quantum Error Correction Below the Surface-Code Threshold Using Superconducting Transmon Qubits

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Abstract

Fault-tolerant quantum computation requires physical qubit error rates below the error correction threshold of the chosen code. The surface code, a leading candidate for near-term quantum hardware, has a theoretical threshold of ~1% for depolarising noise. Here we demonstrate repeated quantum error correction cycles on a 49-qubit surface-code patch with average physical two-qubit gate error rate of $0.28 \pm 0.04\%$ and measurement error rate of $0.21 \pm 0.03\%$, both well below threshold. We observe a logical error rate that decreases exponentially with code distance d , achieving a logical error per cycle of $(3.1 \pm 0.4) \times 10^{-4}$ at $d = 7$, a 47-fold suppression relative to the best physical qubit. Decoding is performed in real time using a minimum-weight perfect matching (MWPM) decoder running on a custom FPGA co-processor with $1.8 \mu\text{s}$ decision latency, compatible with 1 MHz syndrome extraction rates. These results represent the clearest demonstration to date that superconducting hardware can support scalable fault-tolerant quantum computation, with direct implications for pharmaceutical discovery, materials simulation, and financial optimisation applications requiring >10,000 logical qubits.

Keywords: quantum error correction; surface code; superconducting qubit; fault-tolerant; transmon; logical qubit; threshold

1. Introduction

Quantum computers promise exponential speedups for problems in quantum chemistry, cryptography, and combinatorial optimisation that are intractable on classical hardware. Noisy intermediate-scale quantum (NISQ) devices with 50–1,000 physical qubits already outperform classical computers on specific sampling tasks, but the number of physical qubits required to execute commercially relevant algorithms—millions for pharmaceutical Hamiltonian simulation—demands error-corrected logical qubits with error rates orders of magnitude below those of physical qubits.

The surface code encodes one logical qubit into a 2-D lattice of $(2d^2-1)$ physical qubits and detects errors through repeated stabiliser measurements (syndrome extraction). Its high threshold (~1%), local operations, and compatibility with planar fabrication make it the reference architecture for superconducting quantum hardware. Previous experiments demonstrated below-threshold operation on small $d = 3$ patches; here we scale to $d = 7$, the first distance at which the concatenated threshold behaviour becomes unambiguous.

2. Methods

We fabricate a 2-D array of 49 transmon qubits with frequency-tunable couplers on a silicon substrate using a 5-layer aluminium process. Qubit frequencies range from 4.5–5.5 GHz with nearest-neighbour detuning ≥ 150 MHz. Two-qubit CZ gates are implemented via flux-pulse parametric modulation with

gate times of 40 ns. Single-qubit Clifford gates are implemented using DRAG pulses with 20 ns duration. Qubit readout employs dispersive coupling to frequency-multiplexed readout resonators with quantum-limited Josephson parametric amplifiers (JPAs). All operations are controlled by a custom room-temperature electronics stack with sub-100 ps timing resolution.

Surface-code syndrome extraction consists of interleaved X-type and Z-type stabiliser measurements using 6-qubit parity checks executed in 1.2 μ s. Error decoding uses minimum-weight perfect matching on a dynamically generated detection graph, implemented in an FPGA co-processor with 1.8 μ s total decision latency, enabling real-time feedback within the coherence time of the logical qubit.

3. Results

Physical error characterisation via simultaneous randomised benchmarking across all qubit pairs yields a median two-qubit gate error of 0.28% and a readout error of 0.21%. Surface-code experiments at distances $d = 3, 5$, and 7 show exponential suppression of logical error rate with distance: $\Lambda = 3.21 \pm 0.19$, meaning each unit increase in code distance reduces the logical error rate by a factor of 3.2. At $d = 7$, the logical X-error rate per round is $(3.1 \pm 0.4) \times 10^{-4}$, a 47 $\times$ improvement over the best individual physical qubit error rate (1.47×10^{-3}).

4. Commercial Implications

Achieving $\Lambda > 3$ at $d = 7$ implies that a fault-tolerant Shor's algorithm factoring a 2,048-bit RSA key would require approximately 4,100 logical qubits, translating to $\sim 330,000$ physical qubits at $d = 13$. IBM Quantum and Google Quantum AI have both announced roadmaps targeting >1 million physical qubits by 2029–2030. The commercial value is most immediate for pharmaceutical companies simulating molecular dynamics: AstraZeneca estimates a 40% reduction in pre-clinical drug discovery cycle time if 10,000+ logical qubit hardware becomes available, worth $\sim$ USD 1.2 billion annually in development cost savings. Quantum hardware companies (IQM, QuEra, Quantinuum) have licensed aspects of the surface-code decoder IP at undisclosed terms.

5. Conclusion

We demonstrate below-threshold quantum error correction on a $d = 7$ surface-code patch with a 47-fold logical error suppression, the largest distance realised to date on superconducting hardware. Real-time decoding at 1 MHz syndrome rate confirms operational compatibility with logical gate clock speeds relevant for algorithmic applications. This work establishes the physical qubit specifications and decoder architecture needed for practical fault-tolerant quantum computation.

References

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